

RYAN MITCHELLIN

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Career Objective

Recent Computer Science graduate with a Mathematics minor and SOA Exams P and FM, offering a strong blend of programming, analytical, and quantitative skills. Open to opportunities in software development, data analytics, actuarial science, or related fields where I can contribute to solving complex problems through technology and data.

Technical Skills

Languages: Python, SQL, Java, C++, JavaScript, TypeScript, C, HTML, CSS, MATLAB, x86 Assembly

Frameworks & Libraries: Pytorch, Tensorflow, Keras, Flask, Django, ReactJS, Angular, JUnit, JaCoCo, Maven, AWT, JSON, Git

Developer Tools: GitHub, VS Code, Jupyter, Tableau, Excel-VBA, MySQL, SQLite, PostgreSQL, IntelliJ IDEA, Linux, macOS, MS Office

Mathematical & Analytical Tools: Machine Learning, Data Analysis, Regression & Statistical Modeling, Risk Modeling, Probability Theory, Financial Mathematics, Linear Algebra, Numerical Analysis, Calculus I-III, Discrete Mathematics, Financial Accounting

Projects

DermaVision AI | Python, Tensorflow, Keras, Javascript, Flask

- Collaborated with a team of 5 people to develop an image classification model using Convolutional Neural Networks with cross-validation and hyperparameter tuning via random search to classify skin lesion images for training, validation, testing.
- Deployed the trained Keras model into a web application using Flask to provide real-time probability predictions, enabling practical decision-support for classification and predictive modeling tasks.

AI Youtube Shorts Generator | Python, Groq API, Cloudflare FLUX API, Edge TTS

- Developed an end-to-end text-to-video pipeline that converts titles and scripts into vertical YouTube Shorts using AI-generated narration, images, subtitles and video composition.
- Integrated multiple AI APIs and media-processing tools to automate content generation, synchronize narration with subtitles, and produce ready-to-upload short-form videos efficiently.

Chess Prediction Optimization | Python, Numpy, Matplotlib

- Analyzed player win probabilities using ELO difference and Bayesian Inference with Maximum a Posteriori (MAP) estimation. Calculated prior/posterior distributions to evaluate model performance - mirroring techniques used in risk modeling.
- Explored non-linear optimization via sigmoid cross-entropy loss, implementing vanilla gradient descent and Adam optimizer, enhancing statistical models for predictive analysis, risk estimation and performance optimization.

Reporting Website | TypeScript, HTML/CSS, Angular, Bootstrap, Django, SQLite

- Designed a dynamic website using Angular for reporting suspicious activities, utilizing components, directives, pipes, and services for an intuitive user interface and efficient data flow between the user and the server.
- Integrated the Leaflet API visualizing reported incidents on map, with data dynamically retrieved and managed via Django backend using SQLite database, enhancing data visualization and user engagement.

Education

Computing Science Major and Mathematics Minor, Simon Fraser University (SFU) Sep. 2022-Apr. 2026
Bachelor of Science (CGPA: 3.59/4.33) Burnaby, BC

SOA/CAS Exams Exam FM - Passed (Aug. 2025), Exam P - Passed (Jul. 2025), Exam SRM - Scheduled (Oct. 2026)

Fraser International College (FIC) Sep. 2021-Sep. 2022
University Transfer Program II: Computing Science (CGPA: 3.77/4.33) Burnaby, BC

Work Experience

Operations & Service Associate Jan. 2022-Present
Compass Group Canada Burnaby, BC

- Supported daily operations by coordinating tasks while maintaining accuracy, efficiency and service standards.
- Collaborated with team members to improve workflows and deliver consistent service in fast-paced environments.

Teaching Assistant May. 2022-Apr. 2023
Kumon Math and Reading Centre Burnaby, BC

- Guided elementary and secondary students on math concepts and problem-solving techniques to enhance their learning.
- Assessed homework and classwork accurately to reflect areas of strength and weakness in students' learning, maintaining accurate student progress records in the database.

Volunteering Experience

Manager Jan. 2023-Dec. 2023
Peer Education, FIC Burnaby, BC

- Managed a team of 30 students who supported approximately 300 students per semester across multiple courses through peer tutoring and academic guidance.
- Hosted workshops, conducted member interviews, and organized leadership training days to prepare new tutors and mentors for their roles.